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ORIGINAL PAPER

Structure and dynamics of anacardic acid in hexane, ethanol, and carbon tetrachloride: a molecular dynamics investigation

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Abstract

Context Anacardic acid (AA), a key compound in cashew nut shell liquid, is used in medicines and food preservation because of its antimicrobial and antioxidant properties. AA has four forms: saturated, monoene, diene, and triene. Extracting these forms using different solvents is difficult through experiments. To solve this, molecular dynamics (MD) simulations are used to study how AA behaves in three solvents: hexane, ethanol, and carbon tetrachloride. The results show that ethanol forms stronger hydrogen bonds with AA and allows higher movement of AA molecules, making it a better solvent for extraction. These findings help in selecting efficient and sustainable solvents for AA extraction.

Methods MD simulations utilize the Optimized Potential for Liquid Simulations force field to describe the interactions of AA with hexane, ethanol, and carbon tetrachloride. MD simulations are performed using GROMACS open-source package. Structural properties, such as radial distribution functions and hydrogen bonding, and transport properties, like mean square displacement (MSD), are studied to understand how AA behaves in each solvent. These simulations reveal detailed interactions between AA and the solvents, showing why ethanol works better for extraction.

Keywords Molecular dynamics simulation · Anacardic acid · Solvent extraction · Structural properties · Transport properties · Hexane · Ethanol · Carbon tetrachloride

Introduction

Cashew Tree species Anacardium occidental, the most recognized and cultivated species of the group, grows very well in nutrient-rich soil [1]. Cashew nut shell liquid (CNSL) is a co-product obtained from the processing companies of cashew nut shell along with the cashew nut meal, testa shell, and cashew nut shell [2]. CNSL from cashew tree Anacardium occidentale majorly consists of anacardic acid (AA) 64% followed by cardanol 11.31%, cardol 2.04%, and 2-methyl cardanol 1.2% [3]. CNSL has independent utility in fungicidal properties [4], antimicrobial [5], antioxidant [6], insecticidal properties (to control and eliminate insects) [7], antifungal [8], and anticancer [8]. The major component of CNSL is AA, obtained from the extraction of natural CNSL and decarboxylated to cardanol due to the heat treatment in the roasting process [9]. Moreover, AA further has four components, namely saturated, monoene, diene, and triene [9]. The AA obtained from CNSL can be used in traditional applications like antimicrobial and antioxidant agents [10] as well as AA-polymer composites can be applied for food-packaging applications [11]. The reported composite of AA-cassava starch proves to have a reduced biodegradation rate without pH impact and thus enhances electroactivity in the composite films [12].

There are several challenges observed during the extraction process. Traditional methods like hot oil and roasting techniques involve high temperatures leading to the decarboxylation of anacardic acid into cardanol, altering the CNSL composition and adversely affecting its quality and color [13]. Thermal methods can also cause CNSL polymerization, further affecting its quality and color negatively as naturally obtained CNSL and anacardic acid possess a dark brown color [14]. The use of organic solvents in extraction is economical, but removing them entirely from the end products is challenging leading to the rise of environmental concern. A list of solvents like methanol [13], ethanol [15], methyl isobutyl ketone [16], petroleum ether [15], carbon

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tetrachloride [17], and supercritical CO_2 [14] are reported for the AA extraction from CNSL. Methanol-mediated extraction shows a decreasing yield of AA as the proportion of methanol increases, and an optimal temperature is necessary to prevent the formation of hard lumps [13]. The yield of AA increases with the increment of pressure for the supercritical CO_2 extraction [14]. Polar solvents like methyl isobutyl ketone, ethanol, carbon tetrachloride, and petroleum ether produce more extraction efficiency than non-polar solvents like hexane [14–18]. The yield of CNSL extracted from cashew nut shells using ethanol using 1:3, 1:4, and 1:5 molar ratios of CNSL to solvent is 32%, 35%, and 37%, respectively [15]. Optimal solvent selection is crucial for efficient extraction, but achieving the complete elimination of colored impurities remains difficult, and hence, optimization becomes an essential criterion for better yield [16]. Additionally, the complexity associated with the extraction, including the creation of stable salts and separating cardanol and cardol from AA, necessitates looking into the molecular level interactions happening between solvent and AA moieties [16]. Such investigation will lead to a detailed understanding of the behaviour of AA in organic solvents and will lead to efficient extraction.

Molecular dynamics (MD) simulation plays a crucial role in food biomacromolecule research, offering dynamic insights into structural changes at a picosecond level [19]. Atomistic molecular dynamics simulation has been used to study proteins in food systems like in soyabean [20], chicken meat breast [21], kiwifruit [18], hazelnuts [22], egg whites [23], cow's milk [24], and peanuts [25]. In the food technology domain, MD simulation is primarily used to study the interactions among food proteins and other compounds, offering insights into molecular-level structural changes. Current literature suggests that MD visualization models may enhance understanding of finer molecular interactions, aiding research in food protein functionality, bioactive substance delivery, processing, storage, and additives [18]. However, MD simulations face limitations in accurately describing random motions and complex collisions in food systems, particularly in semisolid or solid states and challenges also arise in modelling the diverse structures and multicomponent nature of food matrices, necessitating caution and validation with experimental data [19]. However, MD simulation is presumed to offer a powerful computational approach to overcome experimental limitations in studying the extraction of AA from CNSL and understanding its behaviour in different solvents, thereby elucidating the solvent's suitability for extraction. Additionally, MD simulations can elucidate the structural behaviour of AA in different solvent environments, providing valuable information on its conformational preferences, solvation dynamics, and intermolecular interactions. Moreover, MD simulations can assess the transport properties of AA, such as diffusion coefficients and mean square displacement, offering insights into its mobility and transport mechanisms within solvents. These simulations can help predict the efficiency of solvent extraction processes and optimize conditions for maximum yield. Hence, it serves as a powerful tool to mitigate experimental limitations and enhance our understanding of the extraction and behaviour of AA in various solvent environments.

In this investigation, MD simulations are conducted to investigate the behaviour of AA in three different solvents: hexane, ethanol, and carbon tetrachloride. The dynamics of AA in nonpolar and polar solvents delve into the interactions between AA and each solvent. The analysis of trajectories provides insights into the comparative structural arrangement, solvation dynamics, and intermolecular interactions of AA within each solvent environment. This elucidation offers valuable insights into solvent–solute interactions and the suitability of solvents in anacardic acid extraction processes.

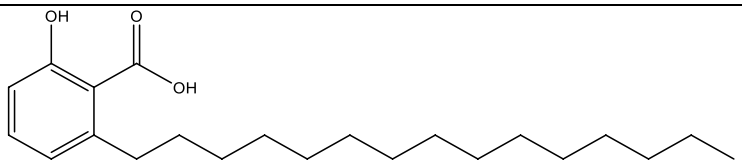
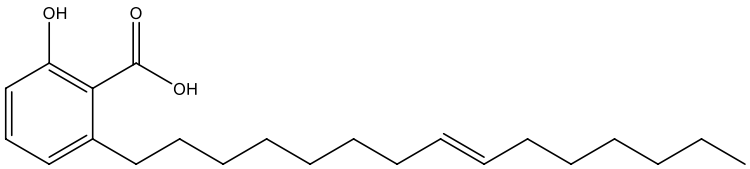
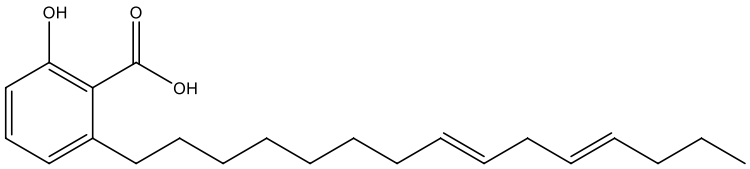
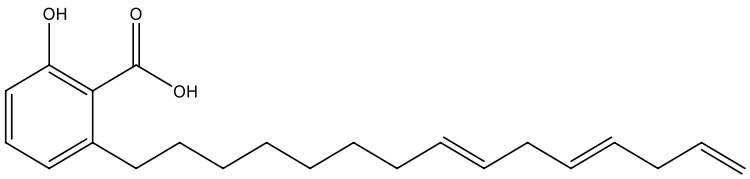
Computational methodology

The AA is composed of saturated, monoene, diene, and triene whose representative structures along with the compositions are given in Table 1 [9]. There are three solvents, namely hexane, ethanol, and CCl_4 being considered for the MD simulations. Table 2 depicts the solvent to solute molar ratio, initial box dimensions for MD simulation, and the number of molecules considered for each system. After generating the initial structures of system components using the Avogadro package [26], the bond, angular, dihedral, improper, and nonbonded interactions, which consist of van der Waals (vdW) and coulombic interactions, are represented through the OPLS force field. The AM1-BCC method is employed to estimate the partial charges on each atom [27]. The function form of the OPLS force field is described in the literature [28, 29].

Framework for MD simulation

The all-atom MD simulation is performed using the GROMACS open-source package [30]. Energy minimization and equilibration followed by the production run create the dynamics of atoms in MD simulation. Energy minimization uses the steepest descent optimization algorithm, with the convergence threshold of 1000 kJ mol^{-1} nm^{-1} which is executed for 500,000 steps with a step size of 0.001 nm. To compute the vdW interaction, a cut-off distance of 1.0 nm is used, and a cut-off distance of 1.2 nm is used to compute long-range electrostatic interactions. The Particle Mesh Ewald (PME) method is used to estimate the long-range electrostatic interaction [31]. A Leapfrog integrator equipped with a 2 fs time step has been employed to integrate all the equations of motion

Table 1 Structures and compositions of AA-saturated, AA-monoene, AA-diene, and AA-triene

AA	Composition (wt%)	Structure
AA saturated	4.3	
AA monoene	33.4	
AA diene	19.9	
AA triene	42.4	

[29]. Equilibration run is further executed with the canonical (NVT) ensemble and the isobaric isothermal (NPT) ensemble for 1 ns and 5 ns, respectively. The NVT ensemble is executed at 300 K constant temperature. The NPT ensemble is executed at 300 K constant temperature and 1.013 bar constant pressure. To maintain the temperature throughout the system in both the equilibration and final production run, a Velocity-Rescale thermostat [31] is used. The pressure is maintained using the Parinello-Rahman barostat [32]. Linear constraint solver (LINCS) algorithm [33] is used to constrain the hydrogen bonds during the equilibration phase. A 25 ns final production run is executed at 300 K temperature and 1.013 bar pressure. The NPT ensemble is considered for the production run. The dynamics of the systems are saved at 10 ps intervals. The trajectories of the atoms of the system are visualized using Visual Molecular Dynamics (VMD) package [34] and the UCSF Chimerax package [35, 36]. Figure 1 represents the initial and final snapshots of the system after the scheduled production run. Figures S1-S3 of the supplementary materials (SM) show the dynamics of different solvent systems. Each MD simulation has been triplicated to address the conformational changes in preparing the systems and thus the analysis results are average of the dynamics.

Analysis of trajectories

Radial distribution function (RDF) and coordination number (*CN*) are the two primary descriptors to establish the distribution of functional moieties of AA in various solvents. The distribution of a moiety with respect to a reference has been described by RDF whose functional form is given below.

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 dr}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta\left(\left|r_i - r_j\right| - r\right) \quad (1)$$

$g(r)$ is the RDF, N is the total number of atoms of the dynamic system, the overall volume of the simulation box is represented by V, the distance between the targeted molecule and other molecules is represented as r, and the width of radial bin is dr. If $z = 0$, then the Dirac delta function which is represented by $\delta(z)$ is equal to 1. The distance between the

Table 2 Specifications of the system

System abbreviation	Components	Number of atoms					Total number of atoms	Initial box geometry (nm^3)	Final box geometry (nm^3)	Final density ($Kg.m^{-3}$)
		Saturated	Monoene	Diene	Triene	Solvent				
AA-Hex	Solute: AA Solvent: Hexane	671	5192	3078	6050	15,000	29,991	10×10×10	6.78×6.78×6.78	828.86
AA-Eth	Solute: AA Solvent: Ethanol	671	5192	3078	6050	15,003	29,994	10×10×10	6.77×6.77×6.77	895.36
AA-CCl_4	Solute: AA Solvent: CCl_4	671	5192	3078	6050	14,996	29,987	10×10×10	8.62×8.62×8.62	1431.06

molecules i and j is $\left|r_i - r_j\right|$. The number of molecules surrounding various solvation shells is given by coordination number (CN) whose mathematical form is given below.

$$CN(r) = 4\pi\rho N \int_o^r g(r) r^2 dr \tag{2}$$

The coordination number is represented by $CN(r)$, ρ represents the particle density, the total number of atoms in the dynamic system is represented by N, $g(r)$ represents the RDF, and the spherical shell radius is r. The integration over the RDF and the radius of the spherical shell counts the number of molecules within a targeted molecule.

Hydrogen bonding (H-bonding) can be assessed by analyzing the distances and angles between hydrogen and electronegative atoms over the course of the simulation. For H-bond formation, the distance between the atoms should be less than equal to 3.5Å and the angle between the atoms should not be more than 30°. H-bond is formed between the hydrogen atoms of the solute AA and solvents hexane, ethanol, and carbon tetrachloride atoms. Equation 3 estimates the number of H-bonds.

$$C_{HB} = \frac{< H(\tau)H(\tau + t) >}{< H(\tau)H(\tau) >} \tag{3}$$

Here, the number of hydrogen bonds is represented by C_{HB}. $H(\tau)$ and $H(\tau + t)$ represent the H-bond order parameter at time τ and $(\tau + t)$, respectively. $H(t)$ is either 0 or 1 depending upon whether the selected pair of molecules develops hydrogen bonds at a specific interval of time.

The MSD analysis involves studying the average squared displacement of anacardic acid molecules over time in each solvent. It provides the insight into the mobility and diffusion behaviour of molecules within a system. By calculating the MSD for anacardic acid in hexane, ethanol, and carbon tetrachloride, it can be assessed that the molecules have moved from their initial positions in each solvent.

$$MSD(t) = < (r_i(t). - r_i(0))^2 > . = \frac{1}{N} \sum_{i=1}^{N} \left[(r_i(t) - r_i(0))^2\right] \tag{4}$$

$r_i(t)$ indicates the location of ^{ith} atom at a time interval of t, and the initial position is represented by $r_i(0)$. According to the Einstein relationship, the gradient of MSD can be used for the calculation of diffusion coefficient which is represented by D.

$$D = \frac{1}{6} \lim_{t \to \infty} \frac{d(MSD(t))}{dt}$$

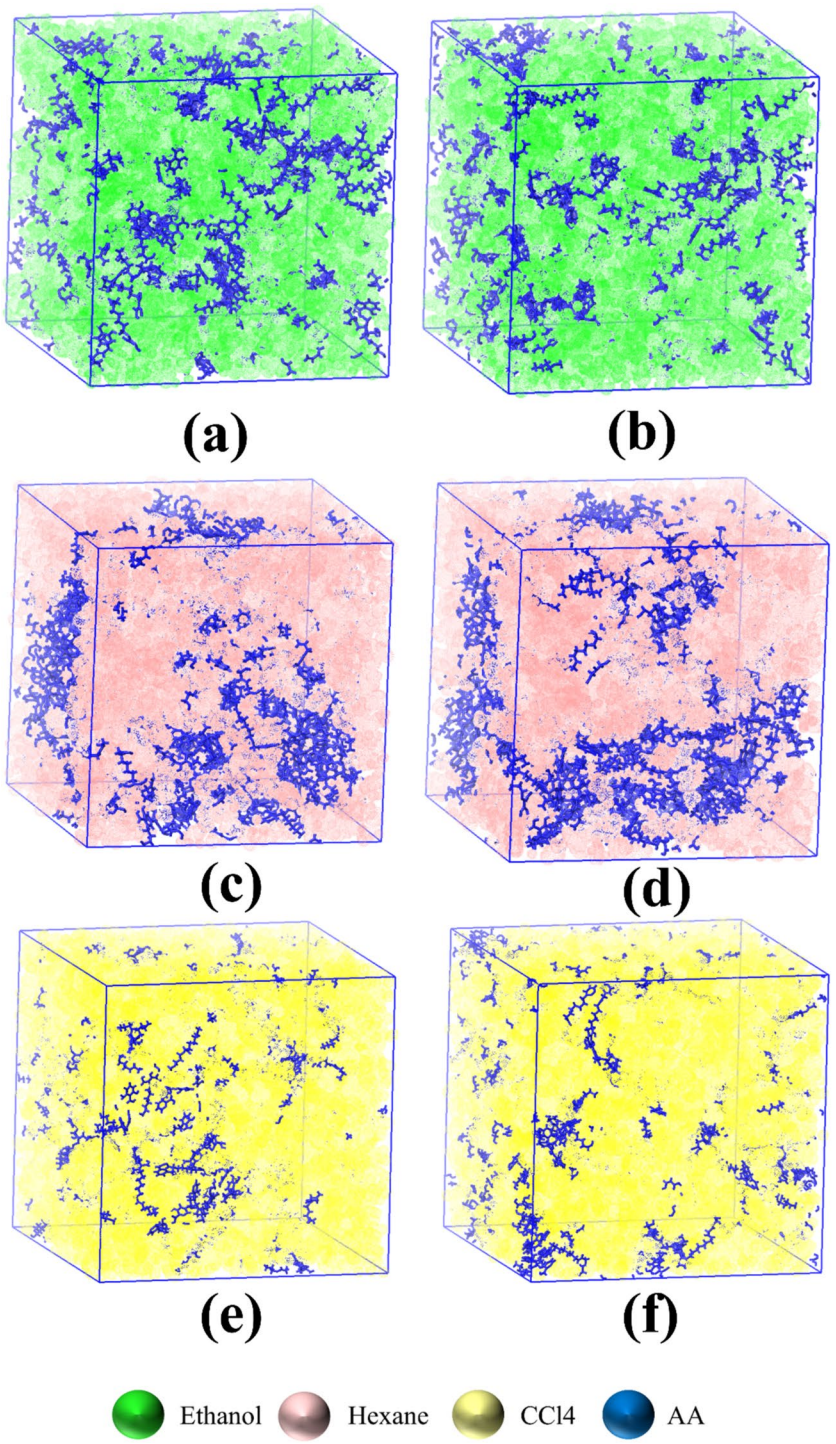


Fig. 1 Initial (**a**, **c**, **e**) and final (**b**, **d**, **f**) snapshots of the dynamic system of AA and ethanol, hexane, and CCl_4, respectively

$$= \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\sum_{i=1}^{N}\left[\left(r_i(t) - r_i(0)\right)^2\right] \quad (5)$$

MSD analysis reveals three distinct behaviors of atoms within the dynamic system: diffusive behaviour (denoted by $\beta = 1$), characterized by random and uncorrelated motion; subdiffusive behaviour (denoted by $\beta < 1$), where atoms experience restriction or hindrance possibly due to intermolecular interactions; and superdiffusive behaviour (denoted by $\beta > 1$), involving accelerated movement and extensive displacement of atoms. The expression of β is given by the following:

$$\beta = \lim_{t\to\infty} \frac{d\{\log(\Delta r^2)\}}{d\{\log(t)\}} \quad (6)$$

Results and discussion

Analysis of distribution of AA in solvent

The orientation of various functional moieties of AA-saturated, AA-monoene, AA-diene, and AA-triene in solvents are represented by RDF and *CN*. Analyzing the molecular structure of AA, it is presumed that the oxygen atom of AA will participate in the meaningful nonbonded interaction. Figures 2 and 3 demonstrate the RDF of AA-saturated, AA-monoene, AA-diene, and AA-triene in ethanol whereas the RDF of various forms of AA in hexane and CCl_4 are given in Figs. S4-S7 and S8-S11 of supplementary material, respectively. Figure 2 illustrates the correlation between the phenolic group oxygen of AA-saturated, AA-monoene, AA-diene, and AA-triene and the hydrogen atom of the alcohol moiety of ethanol solvent. For all the cases, notably, the spread of the first solvation shell is up to 4.0Å, indicating that the encapsulation of all four variants of AA by ethanol has been equally prioritized. However, the peak value of RDF in the first solvation shell marginally varies leading to the highest *CN* of 0.59 for AA-monoene and the lowest

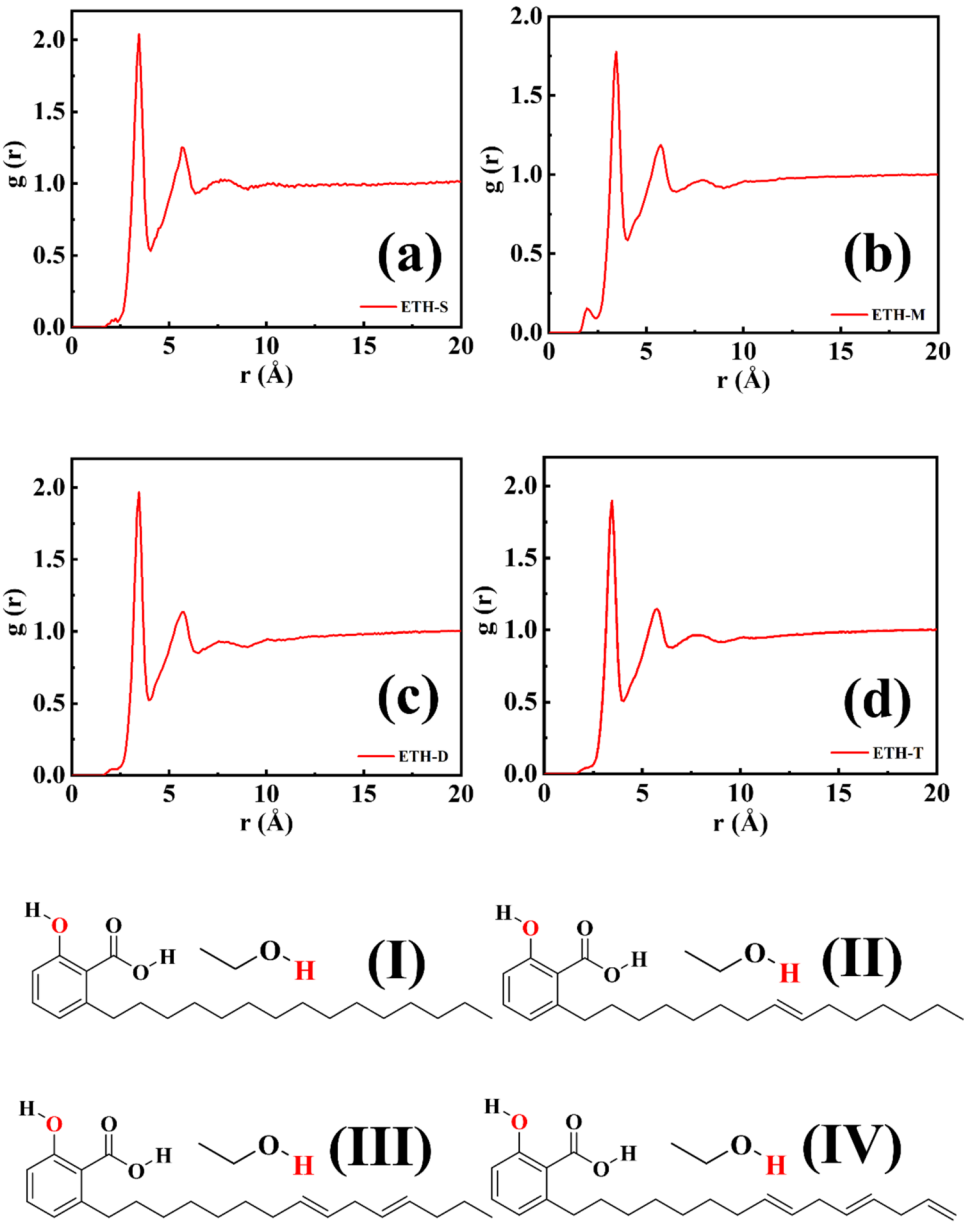


Fig. 2 RDF of oxygen atom of phenolic group of **a** AA-saturated, **b** AA-monoene, **c** AA-diene, and **d** AA-triene with hydrogen atom of alcohol group of ethanol. Schematic diagrams of **I** AA-saturated and ethanol, **II** AA-monoene and ethanol, **III** AA-diene and ethanol, and **IV** AA-triene and ethanol

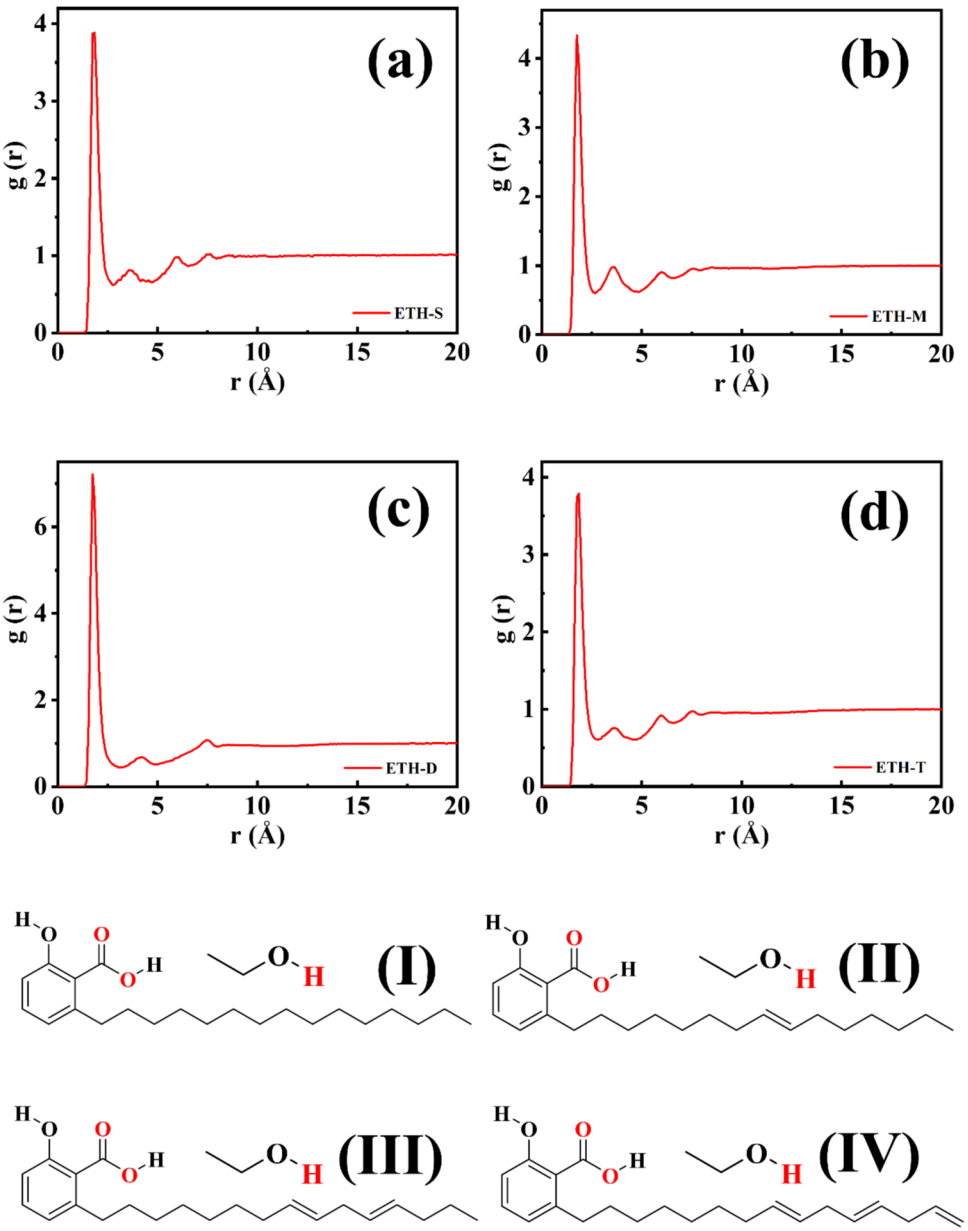


Fig. 3 RDF of oxygen atom of carboxylic acid group of **a** AA-saturated, **b** AA-monoene, **c** AA-diene, and **d** AA-triene with hydrogen atom of alcohol group of ethanol. Schematic diagrams of **I** AA-saturated and ethanol, **II** AA-monoene and ethanol, **III** AA-diene and ethanol, and **IV** AA-triene and ethanol

CN of 0.51 for AA-triene. This marginal variation confirms that ethanol can form a H-bond with all four variants of AA in similar intensity, leading to the simultaneous extraction of saturated, monoene, diene, and triene forms of AA. The spread of the second solvation shell is substantially deemed than the first solvation shell, implying the major role of short-range nonbonded interactions. Figure 3 illustrates the correlation between the carboxylic acid group oxygen of AA-saturated, AA-monoene, AA-diene, and AA-triene and the hydrogen atom of the alcohol group of ethanol, serving as the solvent. The spread of first solvation cells of AA-saturated, AA-monoene, AA-diene, and AA-triene are 2.75Å, 2.75Å, 3.15Å, and 2.75Å, respectively, indicating stronger H-bonding than the previous. The sharpness of RDF in the first solvation shell results higher number of hydroxyl moieties of ethanol oriented towards the carboxylic acid moiety of AA variants. Like the previous, the second solvation shell largely diminishes, emphasizing the dominance of short-range nonbonded interactions over the long-range electrostatic interaction.

Hexane is a nonpolar solvent, and hence, the RDFs of the AA-hexane system between oxygen moieties of AA and hydrogen atoms of hexane (-CH3 and –CH2) do not produce any peak, and the solvation shells are not prominently defined (Figs. S1-S4 of supplementary material). Figures S4 and S5 of supplementary material illustrate the correlation between the phenolic group oxygen and carboxylic group oxygen of AA-saturated, AA-monoene, AA-diene, and

AA-triene, respectively, and the hydrogen atom of hexane solvent. For all the cases, there is no prominent peak and the solvation cells are not prominently defined. There is a little peak at around 4.5Å. These marginal variations in the peaks can be due to weak vdW forces between AA-variants and hexane. However, in Fig. S1, the little peak value of RDF in the first solvation shell varies leading to the highest *CN* of 0.76 for AA-diene and the lowest *CN* of 0.41 for AA-saturated. However, in Fig. S2, the slight peak value of RDF in the first solvation shell varies leading to the highest *CN* of 0.78 for AA-diene and the lowest *CN* of 0.45 for AA-saturated. Figures S6 and S7 of supplementary material illustrate the correlation between the phenolic group oxygen and carboxylic group oxygen of AA-saturated, AA-monoene, AA-diene, and AA-triene, respectively, and the hydrogen atom of hexane solvent. For all the cases, there is no prominent peak and the solvation cells are not prominently defined.

Figures S8 and S9 of SM illustrate the correlation between the phenolic group oxygen and carboxylic group oxygen of AA-saturated, AA-monoene, AA-diene, and AA-triene, respectively, and the chlorine atom of CCl_4 solvent. For all the cases, there is no prominent peak and the solvation cells are not prominently defined. There is a little peak at around 4.0Å. These slight variations in the peaks can be due to weak vdW forces between AA-variants and CCl_4. Figures S10-11 of SM illustrate the correlation between the phenolic group hydrogen and carboxylic group hydrogen of AA-saturated, AA-monoene, AA-diene, and AA-triene, respectively, and the chlorine atom of CCl_4 solvent. For all the cases, there is no prominent peak and the solvation cells are not prominently defined. There is a little peak at around 5.0Å. These marginal variations in the peaks can be due to weak vdW forces between AA-variants and CCl_4. However, in Fig. S10, the little peak value of RDF in the first solvation shell varies leading to the highest *CN* of 5.1 for AA-triene and the lowest *CN* of 2.27 for AA-saturated. However, in Fig. S11, the little peak value of RDF in the first solvation shell varies leading to the highest *CN* of 5.16 for AA-diene and the lowest *CN* of 3.78 for AA-triene.

Evolution of H-bond network

The analysis of H-bonding reveals the stability of hydrogen bonds formed between ethanol and AA variants. H-bonding between ethanol molecules and functional groups present in anacardic acid enhances the solubility of anacardic acid in the ethanol solvent. Figure 4a–d show the hydrogen bonding between the ethanol and AA-saturated, AA-monoene, AA-diene, and AA-triene, respectively. The average number of hydrogen bonds between the ethanol and AA-saturated, AA-monoene, AA-diene, and AA-triene are 18.0, 151.0, 91.0, and 189.0, respectively. The order of average hydrogen

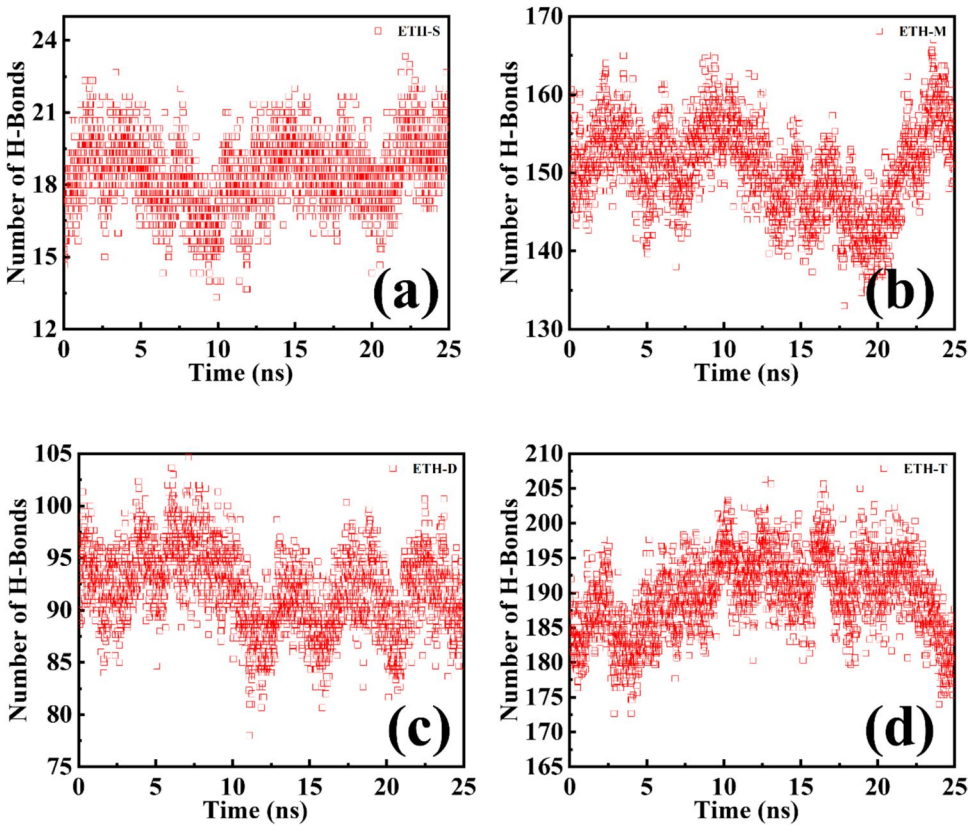


Fig. 4 Hydrogen bonding between ethanol and **a** AA-saturated, **b** AA-monoene, **c** AA-diene, and **d** AA-triene

bonds between the AAs and ethanol is AA-triene > AA-monoene > AA-diene > AA-saturated. The average number of H-bonds becomes proportional with the molar composition of saturate (4%), monoene (33%), diene (20%), and triene (42%). However, large fluctuations are covered for all four cases, implying the dynamic nature of H-bond formation over the course of the simulation. In Fig. 4a, the hydrogen bond count remains stable for the initial 5.0 ns, followed by a gradual decrease to 13 at 9.7 ns. Subsequently, a notable increase to 18 in hydrogen bond count occurs at 22.3 ns. In Fig. 4b, the hydrogen bond count remains stable for the initial 14.0 ns, followed by a gradual decrease of 133 at 19.59 ns. Subsequently, the maxima is observed at 23.9 ns with a count of 167.0 around. Shorter stability of the H-bond is observed for AA-diene, and the maximum count of the H-bond is observed at 7.4 ns. The minimum of 78 is counted at 27.3 ns. However, the most number of H-bond is observed for AA-triene at 16.4 ns, having a count of 206. In Fig. 4d, initially, there is fluctuation in the hydrogen bond count in the system, but after 12.0 ns, there is a notable increase in the hydrogen bond count, peaking at 206 bonds at 16.4 ns. However, the minimum number of H-bonds for AA-triene is 169 which is even greater than the next maximum H-bond formation by AA-dienes. There is no formation of H-bond between AA-variants and hexane, CCl_4 because of the absence of nitrogen, oxygen, and fluorine groups in hexane and CCl_4.

Analysis of dynamics

The dynamics of AA variants in solvents are explored with the time-resolved mean squared displacements. Figure 5 represents the MSD plot of the phenolic group oxygen atom of AA-saturated, AA-monoene, AA-diene, and AA-triene

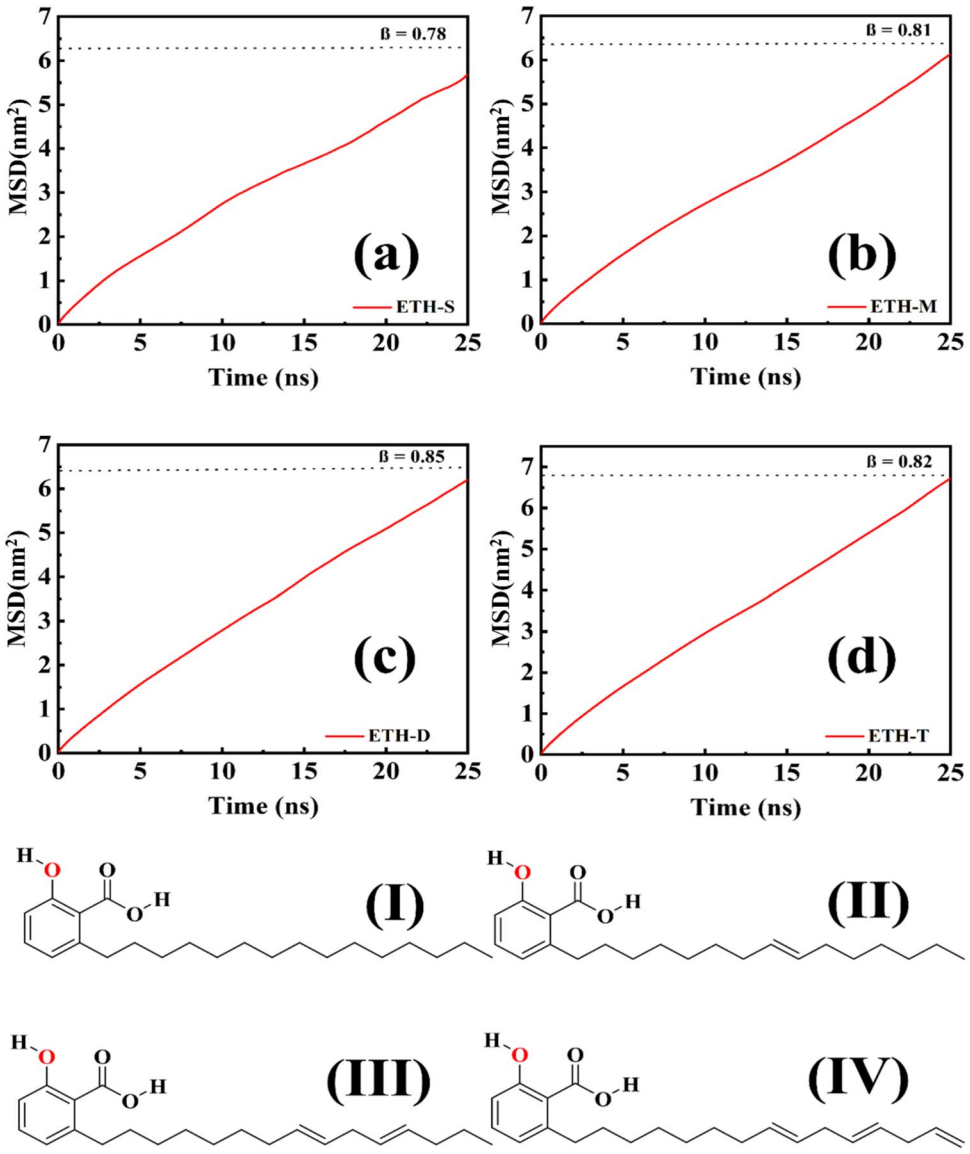


Fig. 5 MSD curve of oxygen atoms of **a** AA-saturated, **b** AA-monoene, **c** AA-diene, and **d** AA-triene in ethanol. Schematic diagrams of **I** AA-saturated, **II** AA-monoene, **III** AA-diene, and **IV** AA-triene in ethanol system

(marked in red) in the ethanol system. All the MSD plots of Fig. 5 reveal only the subdiffusive nature of variants having β values of 0.78, 0.81, 0.85, and 0.82 respectively. Figure 6 represents the MSD plot of the carboxylic acid group oxygen atoms of AA-saturated, AA-monoene, AA-diene, and AA-triene (marked in red). All the MSD plots of Fig. 6 reveal again the subdiffusive nature of movement with β values of 0.79, 0.83, 0.87, and 0.85. There is no particular correlation established with the increment of unsaturation of AA. However, the marginal increment of mobility of carboxylic acid moieties reveals a higher amount of attraction towards the ethanol. The subdiffusive nature suggests that the motion of AA molecules is restricted or hindered.

The MSD plots of AA-variants in hexane are given in Figs. S12 and S13 of supplementary material. Subdiffusive nature of molecular motion of AA prevails in hexane too. However, the mobility becomes slower than in ethanol due to the nonpolar nature of hexane. All the MSD plots of Fig. S12 (phenolic group oxygen in hexane system) of supplementary material reveal only the subdiffusive nature of variants having β values of 0.74, 0.68, 0.68, and 0.63, respectively. All the MSD plots of Fig. S13 (carboxylic acid oxygen in hexane system) reveal again the subdiffusive nature of movement with β values of 0.72, 0.70, 0.65, and 0.66. The MSD plots of AA variants in CCl_4 are given in Figs. S14-S17 of supplementary material. The majorly subdiffusive nature of molecular motion of AA prevails in CCl_4. Due to the nonpolar nature of hexane, the mobility of the AA-variants in hexane is slower as compared to ethanol. Figure S16 represents the MSD plot of the phenolic group hydrogen atom of AA-saturated, AA-monoene, AA-diene, and AA-triene (marked in red) in the CCl_4 system. The possible interaction between hydrogen and chlorine reveals the subdiffusive nature of variants

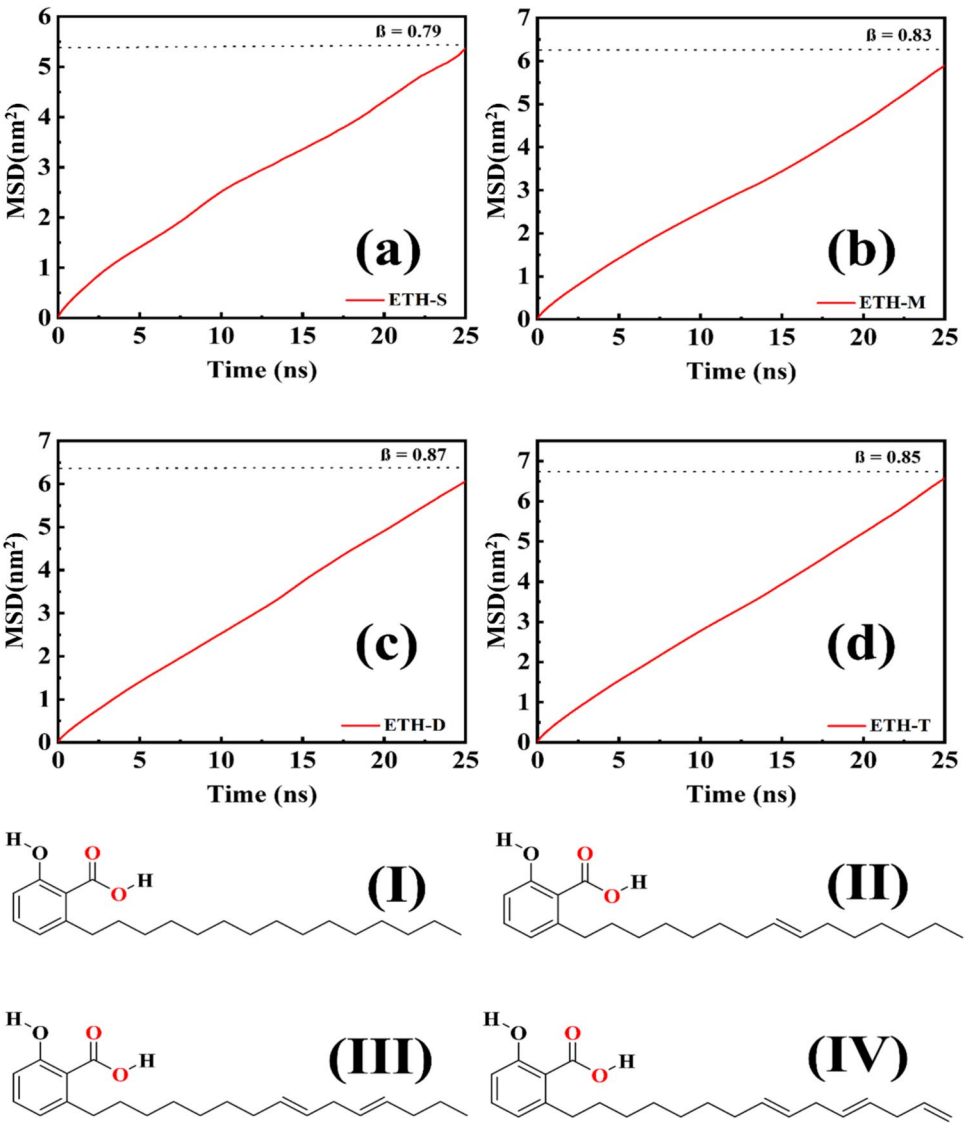


Fig. 6 MSD curve of oxygen atoms of **a** AA-saturated, **b** AA-monoene, **c** AA-diene, and **d** AA-triene in ethanol. Schematic diagrams of **I** AA-saturated, **II** AA-monoene, **III** AA-diene, and **IV** AA-triene in ethanol system

having β values of 0.74, 0.26, 1.02, 0.68, 0.74, 0.29, 0.70, 0.74, 0.46, and 0.99, respectively. Figure S17 represents the MSD plot of the carboxylic acid group hydrogen atoms of AA-saturated, AA-monoene, AA-diene, and AA-triene (marked in red) in the CCl_4 system. Like previous, the subdiffusive nature further prevails with β values of 0.75, 0.64, 0.22, 1.01, 0.83, 0.72, 0.28, 0.74, 0.75, 0.47, and 0.99. The formation of multiple diffusive regimes in Figs. S16 and S17 of supplementary material reveals the relative instability of hydrogen and chlorine interaction.

Conclusion

MD simulation is used to study the behaviour of AA-saturated, AA-monoene, AA-diene, and AA-triene in three different solvents namely hexane, ethanol, and CCl_4. The structural characterization through RDF and *CN* reveals the dominant H-bonding interaction among all variants of AA and ethanol. However, the same analysis nullifies the presence of H-bonding and instead reveals weak vdW interaction between solvent and solute. The order of H-bond between ethanol and AA-triene > AA-monoene > AA-diene > AA-saturated which directly relates to the composition of individual AA-variants in whole AA. Further, the mobility or the motion of molecules of AA-variants is more in ethanol compared to hexane and CCl_4. However, all the variants of AA have shown sub-diffusive behaviour in the three studied solvents indicating the hindered motion of AA-variants This study supports the experimental observation of ethanol as a better solvent over the other two from a molecular standpoint. Ethanol and AA both being polar molecules interact very closely than the nonpolar compounds hexane and CCl_4, and hence, the better yield of AA from CNSL using ethanol is justified.

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Declarations

Competing interests The authors declare no competing interests.

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